A plane wave, described by the given magnetic field, is travelling in a non-magnetic lossless dielectric with dielectric constant of 9.

$$\vec{H} = 0.2\cos(10^9 t - bx - b\sqrt{10}\,z)\ \vec{a}_y \quad (\text{A/m})$$

Find the value of b.
Find the wavelength in the direction of propagation.
Write the corresponding $\vec{E}$.

Consider a plane wave travelling along the positive y axis in a non-magnetic lossy medium with ($\varepsilon_r = 4, \sigma = 0.01\ S/m$). At y=0, the electric field of the wave is given by

$$\vec{E} = 30\cos(10^9\pi\, t + \tfrac{\pi}{4})\ \vec{a}_z \quad (\text{V/m})$$

Find the distance (in m) travelled by the wave to reduce the electric field amplitude to 40% of initial amplitude.
Find the impedance of the medium.
Write the expression for the magnetic field at y=2 m and t=2 ns.

Consider a plane wave propagating in a non-magnetic lossy medium that has loss tangent of 2.4 and ε_r=3.38. The expression of the electric field for this wave is given by

$$\vec{E} = (j\vec{a}_x + 2\vec{a}_y)\cos(10^9 t + \beta z)\, e^{\alpha z} \quad (\text{V/m})$$

Find the wave polarization type (linear, circular, etc.).
Find the complex impedance of the medium.
Write the corresponding instantaneous magnetic field expression.

A circularly polarized uniform plane wave at a frequency of $3 \times 10^9/2\pi$ Hz is traveling in air and is incident normally upon a flat semi-infinite lossy medium with $\varepsilon_r = 9, \mu_r = 1$ and $\sigma = 0.13$ S/m. The incident magnetic field is given by

$$\vec{H} = 2.653\,(j\vec{a}_x - \vec{a}_y)\, e^{-jkz} \quad (\text{mA/m})$$

Find the reflection and transmission coefficients.
Compute the phase and attenuation constants in the lossy medium.
Find the magnitude of the transmitted electric field one meter away from the interface.

A plane wave in air with

$$\vec{H} = 20\, e^{-j(3y+4z)}\ \vec{a}_x \quad \text{mA/m}$$

is incident upon the planar surface of a lossless dielectric material with $\varepsilon_r = 4$ and $\mu_r = 1$ occupying the half-space $z \geq 0$. Produce a sketch of the problem and find

The polarization of the wave (TE/TM).
The angle of incidence.
The frequency of the wave.
The wavelength of the wave inside the dielectric along the propagation direction.
The phasor expression of the reflected electric field.
The instantaneous expression of the transmitted magnetic field.

A uniform plane wave in a lossy nonmagnetic media with $\gamma = 0.2 + j3.4$ /m. The electric field is given by

$$\vec{E} = (5\vec{a}_x + 12\vec{a}_y)\ e^{-\gamma z} \quad (\text{V/m})$$

Compute the magnitude of the electric field at z = 4 m.
Find the loss in dB suffered by the wave in the interval 0 < z < 3 m.